TECHNICAL SUPPLEMENT TO TOPOLOGICAL SEMANTICS FOR SCOPED COMPUTATIONAL PATHS

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Abstract. This supplement retains the full construction proofs, classical examples, formalization maps, and artifact history accompanying the main paper. Its theorem numbering is local to the supplement. References to omitted main-paper sections and results link to that paper. The earlier full manuscript is preserved separately.

S1. Construction calculations

Use the signed-word, endpoint and topology conventions of the main paper. The following proofs fill in its shortened construction lemmas. The main paper retains the explanatory figures.

Lemma S1.1 (continuity of trace realization)**.** *The equal-slot realization map*

$$|\cdot| : \mathsf{Tr}_{\mathcal{P}} \longrightarrow X^I$$

is continuous for the coproduct trace topology.

Proof. On the zero-letter component $W_0 = X$, the realization is the constant-path map. Its adjoint $X \times I \to X$, $(a, t) \mapsto a$, is continuous, so the compact-open exponential law gives continuity into X^I. Fix $n \geq 1$. For each $1 \leq i \leq n$, the closed slot

$$C_i = W_n \times [(i-1)/n, i/n] \subseteq W_n \times I$$

supports the continuous formula

$$(p, t) \longmapsto \widetilde{\rho}(e_i(p))(nt - i + 1),$$

where $e_i : W_n \to \widetilde{E}$ is the ith coordinate. Continuity follows from the continuity of $\widetilde{\rho}$ and of evaluation for the compact-open topology. On a common boundary $t = i/n$, the two formulas agree because composability gives $\widetilde{t}(e_i) = \widetilde{s}(e_{i+1})$. The pasting lemma therefore gives a continuous adjoint $W_n \times I \to X$. Since I is compact Hausdorff, the compact-open exponential law gives a continuous map $W_n \to X^I$. The coproduct universal property now assembles these component maps into the displayed realization map. □

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Lemma S1.2 (continuity of coherent operations)**.** *For either $* = \mathrm{tr}$ or $* = \mathrm{obs}$, let $\iota : X \to T_*$ send a to the coherent identity $(1_a, c_a)$, let $\sigma : T_* \to T_*$ send (p, γ) to (p^{-1}, γ^{-1}), and let*

$$\mu_* : T_*^{(2)} \longrightarrow T_*, \qquad ((p, \gamma), (q, \delta)) \longmapsto (p; q, \gamma \star_{\ell(p),\ell(q)} \delta),$$

where the weighted operation uses the discrete trace lengths. Then all three maps are continuous.

Proof. For T_{obs}, it is enough to check continuity after applying κ_{obs}. The three resulting observable codes are respectively

$$(a, a, 0, c_a, c_a), \quad (t(p), s(p), \ell(p), |p|^{-1}, \gamma^{-1}),$$

and, on the composable pullback,

$$\big(s(p), t(q), \ell(p) + \ell(q), |p| \star_{\ell(p),\ell(q)} |q|, \gamma \star_{\ell(p),\ell(q)} \delta\big).$$

The first is continuous because constant paths depend continuously on a; the second uses reversal on X^I; and the third uses addition on the discrete length coordinate and continuity of the weighted operation on each fixed length component. Each code depends continuously only on the observable codes of the input(s), so the initial-topology criterion proves continuity of all three operations. For T_{tr}, the corresponding first coordinates are the continuous word maps $a \mapsto 1_a$, $p \mapsto p^{-1}$, and $(p, q) \mapsto p; q$ on the trace carrier; the second coordinates are the same continuous path operations just used. The initial-topology criterion for κ_{tr} therefore proves continuity in the trace-sensitive topology as well. Reversal and weighted concatenation preserve the coherence predicate by reversing and concatenating the endpoint-fixed homotopies. The Lean development flattens the parenthesized internal trace to a signed word so that these first-coordinate maps can be checked. □

Lemma S1.3 (weighted concatenation invariance)**.** *Let $\alpha \simeq_{\partial I} \alpha'$ and $\beta \simeq_{\partial I} \beta'$ be endpoint-fixed homotopies of composable paths. Write c_x for the constant path at x, and assume explicitly that*

$$\begin{aligned}
m = 0 &\Rightarrow \alpha(0) = \alpha(1) \text{ and } \alpha \simeq_{\partial I} c_{\alpha(0)},\\
n = 0 &\Rightarrow \beta(0) = \beta(1) \text{ and } \beta \simeq_{\partial I} c_{\beta(0)},\\
m' = 0 &\Rightarrow \alpha'(0) = \alpha'(1) \text{ and } \alpha' \simeq_{\partial I} c_{\alpha'(0)},\\
n' = 0 &\Rightarrow \beta'(0) = \beta'(1) \text{ and } \beta' \simeq_{\partial I} c_{\beta'(0)}.
\end{aligned}$$

For arbitrary $m, n, m', n' \in \mathbb{N}$, with the zero-length conventions above,

$$\alpha \star_{m,n} \beta \simeq_{\partial I} \alpha' \star_{m',n'} \beta'.$$

Proof. When all four weights are positive, let $\lambda_{m,n} : [0, 1] \to [0, 1]$ be the piecewise-linear endpoint-fixing homeomorphism that maps $m/(m + n)$ to $1/2$ and is linear on the two sides. Explicitly,

$$\lambda_{m,n}(t) = \begin{cases} \dfrac{m+n}{2m} t, & 0 \le t \le \dfrac{m}{m+n}, \\[2ex] \dfrac{1}{2} + \dfrac{m+n}{2n}\left(t - \dfrac{m}{m+n}\right), & \dfrac{m}{m+n} \le t \le 1. \end{cases}$$

Then

$$\alpha \star_{m,n} \beta = (\alpha \star_{1,1} \beta) \circ \lambda_{m,n}.$$

The straight-line homotopy $(1-u)\lambda_{m,n} + u\lambda_{m',n'}$ fixes 0 and 1, so changing the breakpoint gives an endpoint-fixed reparametrization homotopy. Horizontal concatenation of the given homotopies at the common breakpoint $1/2$ gives

$$\alpha \star_{1,1} \beta \simeq_{\partial I} \alpha' \star_{1,1} \beta'.$$

Combining these homotopies proves the claim when all weights are positive. If a weight is zero, first use the corresponding zero-slot hypothesis to replace the corresponding factor by the constant path at its endpoint. Such a factor has coinciding endpoints by the displayed hypothesis, so the stated convention then deletes that constant factor. When the zero/nonzero status changes between the two sides, this replacement inserts or removes the factor up to endpoint-fixed homotopy. The remaining positive-weight factors are handled by the preceding reparametrization argument; when both weights are zero, the convention yields the common endpoint constant. Thus all zero-length cases follow as well. □

Lemma S1.4 (backtracking contracts in every space)**.** *For any continuous path $\alpha : I \to X$, the formula*

$$H(u,t) = \alpha(\min\{2t,\, 2(1-t),\, 1-u\}) \qquad (u,t \in I)$$

is an endpoint-fixed homotopy from $\alpha \star_{1,1} \alpha^{-1}$ to $c_{\alpha(0)}$.

Proof. The minimum is a continuous map $I^2 \to I$, so H is continuous. At $u = 0$ its argument is the triangular function $\min\{2t, 2(1-t)\}$, which traverses α and then retraces it. At $u = 1$ it is identically zero, and for every u its values at $t = 0, 1$ are zero. These are the required boundary conditions. Applying the same construction to α^{-1} proves the other inverse law. □

The geometry of reversal, coherence, observational separation and backtracking is illustrated in the main paper's Figures 1–4.

S2. Universal projection lemmas

The universal presentation and its scoped relation are those of the main paper. These projection and fixed-endpoint arguments supplement the full ordinary-composition proof retained there.

Lemma S2.1 (coherent-path quotient identification)**.** *For the universal presentation, the projection*

$$\pi : T_{\mathcal{U}_X} \longrightarrow X^I, \qquad \pi(p,\gamma) = \gamma,$$

is a continuous quotient map. Consequently, the quotient of $T_{\mathcal{U}_X}$ by endpoint-fixed homotopy of the chosen representative is canonically homeomorphic to the usual quotient-topologized path-class space

$$\Pi_1^q(X) = X^I/{\simeq_{\partial I}},$$

with the endpoint maps retained.

Proof. For T_{obs}, continuity of π is immediate from the fifth coordinate of κ_{obs}. For T_{tr}, it is the restriction of the second projection $\mathsf{Tr}_{\mathcal{P}} \times X^I \to X^I$. Every $\gamma \in X^I$ has the coherent representative

$$\sigma(\gamma) = (\langle\gamma\rangle, \gamma),$$

where $\langle\gamma\rangle$ is the one-letter universal trace and the coherence is the constant endpoint-fixed homotopy. The map σ is continuous on the one-letter coproduct

component for T_{tr}; it is also continuous for T_{obs} because its observable code is built continuously from γ. Since $\pi\sigma = \mathrm{id}$, if $\pi^{-1}(U)$ is open then $U = \sigma^{-1}(\pi^{-1}(U))$ is open. Thus π is quotient. The general quotient-factorization lemma applied to the homotopy relation now induces mutually inverse continuous maps between the two displayed quotient spaces. □

Lemma S2.2 (fiberwise coherent-path quotient)**.** *Fix $a, b \in X$, and let*

$$X^I(a,b) = \{\gamma \in X^I : \gamma(0) = a,\ \gamma(1) = b\}$$

carry its subspace topology. For $* = \mathrm{tr}, \mathrm{obs}$, define the coherent carrier with these fixed endpoints

$$T^*_{\mathcal{U}_X}(a,b) = \{(p,\gamma) \in T^*_{\mathcal{U}_X} : s(p) = a,\ t(p) = b\}$$

with its own subspace topology, and define $G^*_{\mathcal{U}_X}(a,b)$ to be its quotient by scoped equivalence, with the quotient topology. The restricted projection

$$\pi^*_{a,b} : T^*_{\mathcal{U}_X}(a,b) \longrightarrow X^I(a,b)$$

is quotient. Hence $G^*_{\mathcal{U}_X}(a,b)$ is canonically homeomorphic to the endpoint-fixed homotopy quotient of $X^I(a,b)$.

Proof. The projection is continuous for either representative topology. Its one-letter section

$$\sigma_{a,b}(\gamma) = (\langle\gamma\rangle, \gamma)$$

lands in the fixed endpoint fiber and is continuous by the same coordinate argument as in Lemma S2.1. If $(\pi^*_{a,b})^{-1}(U)$ is open, then $U = \sigma^{-1}_{a,b}((\pi^*_{a,b})^{-1}(U))$ is open. Thus the restricted projection is quotient without appealing to a restriction theorem for the global projection. In the universal presentation, scoped equivalence on this fiber is exactly endpoint-fixed homotopy, so quotient factorization gives the claimed homeomorphism. □

The corresponding illustration is Figure 9 in the main paper.

S3. Circle, torus and product proofs

These arguments use the finite presentations and endpoint conventions of Section 8 of the main paper. The global based-arrow subspace comparison is established separately from the based-carrier quotient.

Theorem S3.1 (circle winding classification)**.** *The finite-generator scoped presentation is geometrically complete on the based loop fiber, and winding number is a group isomorphism and a homeomorphism*

$$L \cong \mathbb{Z},$$

where $\mathbb{Z}$ is discrete. Moreover, the inverse realization comparison

$$S_0 : L \xrightarrow{\cong} G_{\mathcal{P}}(0,0)$$

is a homeomorphism. In particular the scoped presentation is nondegenerate: the images of the zero and one winding loops are distinct.

Proof. A based loop has a unique lift to $\mathbb{R}$ starting at 0. Its endpoint is an integer, and endpoint-fixed homotopy preserves that integer. Straight-line interpolation of the lift with $t \mapsto tn$ shows that every loop of winding n is homotopic to λ_n. The lift of a concatenation is the first lift followed by a translate of the second, so winding is additive. Thus winding identifies L with $\mathbb{Z}$ as groups.

The finite-generator reduction lemma in the main paper sends every trace to c_n, where n is its winding number, so the normal-form criterion gives geometric completeness. Explicitly, S_0 sends a loop class of winding n to $q(c_n, \lambda_n)$. Every scoped class has this form by normalization, and soundness prevents different windings from giving the same scoped class. Hence S_0 is inverse to realization. In particular, the zero and one winding classes are distinct.

It remains to prove continuity of S_0. Give $\mathbb{T}$ its standard quotient metric. If two based loops γ, δ are uniformly less than $1/2$ apart, their difference has a unique continuous representative $d : I \to (-1/2, 1/2)$, with $d(0) = d(1) = 0$. Then $\gamma(t) + [u\, d(t)]$ joins them relative to endpoints. Compact-open and uniform topologies agree here since I is compact and $\mathbb{T}$ is metric. Winding is therefore locally constant and descends continuously from L to discrete $\mathbb{Z}$. Choosing (c_n, λ_n) for each n gives a continuous map from discrete $\mathbb{Z}$ to the coherent carrier and then to its scoped quotient. This constructs S_0 continuously; projection to the chosen loop supplies its continuous inverse on the fixed-endpoint quotient.

To check the topology inherited by $G_{\mathcal{P}}(0,0)$ from the *global* scoped arrow space, consider a coherent raw representative with source x and chosen geometric loop γ. Every trace in this finite presentation has equal source and target. Define

$$W(x, \gamma) = \mathrm{wind}\big(t \mapsto \gamma(t) - x\big).$$

Translation of a variable-basepoint loop to zero is continuous in the compact-open topology. The observable raw topology includes both x and γ, and winding is locally constant on based loops, so W is continuous on the full raw carrier. Scoped equivalence preserves the source and gives an endpoint-fixed homotopy of the chosen loops. Thus W descends continuously to the global arrow quotient. On the based subspace it is ordinary winding. Normal-form completeness makes this restriction injective, while the classes of (c_n, λ_n) make it surjective. Its inverse is continuous because $\mathbb{Z}$ is discrete. This proves the claimed homeomorphism for the actual subspace topology without restricting a quotient map. The based loop-class quotient is discrete; the loop space itself need not be. □

Theorem S3.2 (torus winding classification)**.** *For the torus $\mathbb{T}^2$, the coordinate winding pair is a group isomorphism and a homeomorphism*

$$L_{\mathbb{T}^2} \cong \mathbb{Z} \times \mathbb{Z},$$

where $\mathbb{Z} \times \mathbb{Z}$ is discrete. The finite-generator scoped presentation is geometrically complete on the based loop fiber. Moreover, the inverse realization comparison

$$S_{\mathbb{T}^2} : L_{\mathbb{T}^2} \xrightarrow{\cong} G_{\mathcal{P}}((0,0),(0,0))$$

is a homeomorphism.

Proof. The universal cover is the product map

$$\mathbb{R}^2 \longrightarrow \mathbb{T}^2, \qquad (x, y) \longmapsto ([x], [y]).$$

The lift of a based loop has endpoint $(m, n) \in \mathbb{Z}^2$, and endpoint-fixed homotopy preserves this pair. Conversely, the two coordinate projections of any based torus loop are circle loops; the circle lifting theorem gives endpoint-fixed homotopies to

the standard loops of windings m and n. Taking their product gives a homotopy to the simultaneous loop $d_{m,n}$; Lemma 8.3 then gives a homotopy to the sequential realization $|a^m; b^n|$. Thus the coordinate winding pair classifies geometric loops, and it is additive because each coordinate winding is. The signed-word reduction in the main paper gives the corresponding scoped derivation, so the normal-form criterion proves geometric completeness. Projection from coherent representatives to their chosen loops induces the continuous realization comparison. The local-constancy argument constructs its inverse continuously: each coordinate winding number is locally constant by the circle argument. The winding pair thus descends continuously to discrete $\mathbb{Z}^2$, and the map $(m, n) \mapsto (a^m; b^n, d_{m,n})$ is continuous because its domain is discrete. Lemma 8.3 ensures that each pair is coherent. Composing these maps with the scoped quotient map gives $S_{\mathbb{T}^2}$. For the topology inherited from the global scoped arrow space, translate each variable-basepoint torus loop γ by its source x and take the two coordinate windings of $t \mapsto \gamma(t) - x$. This gives a continuous $\mathbb{Z}^2$-valued map on the global quotient: translation and winding are continuous, and scoped soundness makes the value independent of the representative. Its restriction to the based subspace is bijective by the normal form above, with continuous inverse from discrete $\mathbb{Z}^2$. This proves the stated subspace homeomorphism. As for the circle, the loop space itself is not asserted to be discrete. □

Theorem S3.3 (product closure of based completeness)**.** *Suppose $\mathcal{P}$ is geometrically complete on the based-loop fiber at $x \in X$ and $\mathcal{Q}$ is geometrically complete on the based-loop fiber at $y \in Y$. Then $\mathcal{P} \boxtimes \mathcal{Q}$ is geometrically complete on the based-loop fiber at (x, y).*

Proof. Induct on a product trace. Identity traces and single letters are already sorted after inserting a unit trace. For a concatenation, first sort both parts, then use the derived trace interchange to move the first part's vertical projection past the second part's horizontal projection. For an inverse, reverse the sorted trace and use interchange once more. Thus every trace rewrites to its horizontal projection followed by its vertical projection. Since the entire trace starts and ends at (x, y), the horizontal projection is a loop at x, lifted at y, and the vertical word is a loop at y, lifted at x.

Take two coherent product loops whose geometric representatives are endpoint-fixed homotopic. Soundness and coherence let us replace each by its sorted realization. Projecting the resulting homotopy to X gives a homotopy between the two horizontal realizations: every vertical letter projects to a constant segment, which can be removed by an endpoint-fixed reparametrization homotopy. Projection to Y gives the corresponding homotopy between vertical realizations. Completeness of $\mathcal{P}$ and $\mathcal{Q}$ supplies scoped derivations between the respective factor words. Lift the first derivation at y and the second at x, then compose them with the two sorting derivations. This is a scoped derivation between the original product words, as required. □

The corresponding illustration is Figure 10 in the main paper.

S4. What the Lean developments check

Current model boundary: 8 October 2026. The manuscript's trace carrier in Section 2 uses flat signed words with equal-duration slots. The earlier Lean modules

use parenthesized traces and recursively apply Mathlib's half-and-half `Path.trans`. Their observable code retains the exact parametrized realization. Adding the flattened signed word to that code does not remove the binary timing coordinate. Consequently, the existing Lean topology theorems concern the binary model. They must not be read as certificates of the equal-slot observable topology without a separate comparison theorem. Endpoint-fixed homotopy of the two realizations is sufficient for semantic soundness, but does not establish such a topological comparison. The binary development and its verified results are retained. A separate implementation now defines `GeometricTrace.flatRealize` on the same endpoint-indexed tree carrier. Each positive factor in a concatenation receives time proportional to its primitive length; zero-length factors are omitted. The theorem `flatRealize_slot` proves, including slot boundaries and reversal, that letter i occupies exactly $[i/n, (i+1)/n]$. The theorem `flatRealize_eq_of_flatWord` proves equality of parametrized realizations for trees with the same signed word. Thus parentheses are invisible to this realization. The theorem `flatRealize_homotopic_binary` supplies the endpoint-fixed homotopy bridge; it does not compare the two observable quotient topologies.

`EqualSlotTopology` defines separate fixed-endpoint and global observation topologies using this exact flat realization, without retaining a binary realization coordinate. The coherent carrier can be reused: `coherent_iff` proves that coherence with the binary and flat paths are equivalent propositions, and `scoped_sound` transfers rewrite soundness. The literal-word bridge now closes the two additional obligations. `LiteralWord.Word.normalization` reconstructs each tree from its composable signed word using only units, associativity, and reversal laws. An independently generated word relation uses adjacent-letter cancellation and the flattened named rules; `WordRwEq.ofTrace_iff` proves that it agrees with tree rewriting. Equality of geometric paths is never introduced as a rewrite rule.

The literal finite strata have $W_0 = X$ and the compatible positive signed tuples with their product-subspace topologies. Harmless Lean universe lifts are removed by explicit homeomorphisms. Canonical equal-slot reconstruction is continuous on every stratum. The coproduct topology is induced by $(s(p), p)$, where the second coordinate is the complete signed word: the source retains the basepoint of an empty word. The source coordinate is redundant on every positive stratum. The theorems `Word.sensitiveTopology_eq_tupleTopology` and `TotalWord.sensitiveTopology_eq_coproductTopology` then identify the full-word refinements with these genuine literal-word topologies. Neither result identifies a binary observable topology with a flat one.

Proposition S4.1 (binary timing can identify observable codes)**.** *There is a three-generator based circle presentation with no named rewrite rules for which the observable quotients of the fixed-base carriers differ: the binary quotient is not T_0, and the equal-slot flat quotient is discrete.*

Proof. Let $\alpha(t) = [t]$ in $\mathbb{R}/\mathbb{Z}$, and let the primitive labels a, b, c realize α, $\beta = \alpha \star \alpha$, and $\chi = \alpha \star \beta$, respectively, with $\star$ denoting ordinary half-and-half concatenation. The binary traces $p = (a; b); b$ and $q = c; (a; a)$ both realize $\chi \star \beta$. They have length three and can choose this same path as coherent representative, so their binary observable codes agree. Signed occurrence count of c is invariant under structural scoped rewrites and has values zero and one on these two traces. Their scoped classes are distinct. Every open set in the induced raw observable topology

contains both chosen representatives or neither; the same holds for their classes by the definition of quotient topology. Hence that quotient is not T_0.

For the flat construction, the six oriented primitive paths have distinct winding numbers $1, 2, 3, -1, -2, -3$. Restriction to each prescribed equal slot therefore recovers the signed letter. At each fixed length there are finitely many words, with distinct realizations in the Hausdorff compact-open path space. This finite image is discrete, so the fiber of each observed word is open. The discrete length coordinate separates the strata. A scoped-class preimage is a union of word fibers, even when the coherent path varies; thus the fixed-base flat quotient is discrete. This last assertion concerns the quotient of the based carrier, not an unproved subspace comparison with the global quotient. □

`BinaryTimingCollision` implements the exact binary collision and proves failure of T_0 for both the fixed-base trace quotient and the global binary scoped quotient. Coherent-representative elimination transfers the fixed-base trace result to the matching coherent quotient. The actual circle winding computation, flat discreteness, and exclusion of every binary/flat observable homeomorphism are checked in `FlatObservableDiscreteness.BinaryCircle`.

Proposition S4.2 (fixed-endpoint coherent-representative elimination)**.** *For a fixed realization model and endpoints, give traces a topology making realization continuous and coherent representatives the topology induced by (p, γ). If scoped equivalence depends only on the trace, forgetting γ induces a homeomorphism between the coherent quotient and the trace-only scoped quotient.*

Proof. The projection $r(p, \gamma) = p$ is continuous, and $s(p) = (p, |p|)$ is a continuous section. Both maps preserve the quotient relations. We have $rs = \mathrm{id}$, while $sr(p, \gamma)$ is scoped equivalent to (p, γ) because the trace is unchanged. Quotient descent therefore gives continuous mutually inverse maps. No geometric completeness hypothesis is needed. Apply the statement separately to each realization model and each chosen trace topology. □

`CoherentRepresentativeElimination` implements this result for fixed endpoints with the binary observable trace topology and its full-word refinement. Its general declaration accepts any trace topology for which the same binary realization is continuous. `EqualSlotTopology` proves the matching result with the equal-slot canonical section, including the observable trace topology and its full-word refinement. These are separate fixed-endpoint homeomorphisms. `LiteralWord.TotalWord` extends matching equal-slot elimination and structural reconstruction to global arrows, and `LiteralWord.WordPair` extends them to the quotient of the actual endpoint-matched representative subspace. Both the observable and full-word versions are checked; the full-word version is also identified with the genuine literal-word coproduct. These results do not identify the ordinary pullback topology, a based subspace inside a global quotient, or a binary observable topology with a flat one.

The last row records a remaining coverage boundary. The general binary operations and final-domain theorem are checked. The flat homeomorphisms certify arrow and pair spaces; they do not separately check continuity of the canonical flat operations or their compatibility with those homeomorphisms.

Result	Exact local declaration	Assumptions	Model
Binary elimination	`quotientHomeomorph`	Fixed endpoints; realization continuous for supplied trace topology; trace-only scoped relation	Binary
Timing collision	`realize_eq`	Any based path; three discrete labels with the stated realizations	Binary
Based failure of T_0	`fixed_not_t0`	Same three-label system; no named rewrite rules	Binary
Global failure of T_0	`not_t0`	Same three-label system; no named rewrite rules	Binary
Exact slot formula	`flatRealize_slot`	Endpoint-indexed trace; index $i < n$ and $t \in I$	Flat tree
Word independence	`flatRealize_eq_of_flatWord`	Fixed endpoints; equal signed words	Flat tree
Homotopy bridge	`flatRealize_homotopic_binary`	Any endpoint-indexed trace	Both
Flat elimination	`quotientHomeomorph`	Fixed endpoints; flat realization continuous for supplied trace topology	Flat tree
Flat discreteness	`BinaryCircle.flat_traceQuotient_discrete`	Circle paths with six distinct oriented realizations; fixed-base carrier	Flat
General operations and final multiplication	Mathematical proof; no separate flat declaration	Continuous presentation; main Lemma 2.10 and Theorem 3.2	Flat

TABLE 1. Model-boundary verification map. The first row is in `CoherentRepresentativeElimination`; the next three are in `BinaryTimingCollision`, all under `ComputationalPaths.Path.GeometricTopology`. The next three are in `GeometricTrace`, and flat elimination is in `EqualSlotTopology`. The flat-discreteness row is in `FlatObservableDiscreteness`. The earlier eight formal rows passed Lean 4.32.0 kernel checks at source commit `404ac01`, with Mathlib commit `81a5d257`; only `propext`, `Classical.choice`, and `Quot.sound` occur in their axiom closures. Flat discreteness and the nonhomeomorphism conclusion are now formalized in `FlatObservableDiscreteness.BinaryCircle`, using winding numbers of the six actual oriented primitive paths. The earlier manifest `model-boundary-replay.json` at commit `131e4bf` selected 19 actual declarations for direct NanoDa replay, accepted in the exact-head Quality run. The five earlier Comparator selections retain their own scopes. Later selections extend this coverage as described below.

8 October 2026: first substitution selection. The presentation-change milestone added `ScopedWordInterpretation` and `ContinuousWordInterpretation`. The former checked primitive finite-trace substitutions, named-rule-to-scoped preservation, primitive roundtrip extension, and the algebraic `quotientEquiv`. The latter checked `WordSubstitution.mapTrace_length`, `mapTrace_realize`, and `continuous_binary`.

It derived `ScopedWordInterpretation.quotientHomeomorph` for supplied trace topologies and `binaryQuotientHomeomorph` under the primitive conditions of Proposition 5.3. `RedundantGeneratorObservable.quotientEquiv` checks the conservative move of Proposition 5.5; `continuous_oldParity`, `new_classes_ne`, `new_inseparable`, and `not_continuous_inverse` check its topological failure for the fixed-endpoint flat-tree observable quotients. All these declarations passed the Lean 4.32.0 kernel at source commit `d604b75`, with the same pinned Mathlib and only the three standard axioms listed in Table 1. The direct-replay manifest at that milestone selected 37 actual declarations. Its evidence applied to that exact published head. That earlier 37-declaration milestone included no general variable-length flat invariance, global presentation equivalence, or final-pair transport. The current milestone adds those results under the primitive-word continuity hypotheses of Theorem 5.4.

8 October 2026: literal words and variable-length transport. The literal-word milestone added the checked results in Table 2. All use Lean 4.32.0 with Mathlib `81a5d257`; their axiom closures contain only `propext`, `Classical.choice`, and `Quot.sound`. The source snapshot is `0346c8e`. The direct replay manifest selects the actual declarations; exact-head hosted acceptance is recorded with the published artifact, separately from local kernel checking.

Table 2: Literal-word verification map. Names are in `ComputationalPaths.Path.GeometricTopology.LiteralWord`. Every row has source commit `0346c8e`.

Result	Exact declaration	Assumptions	Model
Structural normalization	`Word.normalization`	Any continuous scoped presentation; no named rule used	Both
Same-word rewriting	`Word.scopedEq_of_flatWord`	Parallel trees with equal signed words	Both
Independent relation	`WordRwEq.ofTrace_iff`	Adjacent-letter cancellation and flattened named rules	Literal/ tree
Stratum execution	`Strata.continuous_flatRealize_family`	Continuous primitive realization; genuinely composable tuples	Flat
Initial coproduct code	`Strata.totalWordTopology_eq_induced_code`	Empty basepoint retained; compatible finite tuple strata	Flat
Variable-length family	`Strata.continuous_flatRealize_of_flatWord`	Continuous endpoints and full signed word; typed compatibility	Flat
Fixed tuple comparison	`WordRwEq.tupleQuotientHomeomorph`	Matching full-word topology and scoped relation	Flat
Global comparison	`TotalWord.coproductQuotientHomeomorph`	Matching equal-slot coherent carrier and literal coproduct	Flat
Final-pair comparison	`WordPair.coproductQuotientHomeomorph`	Quotient of actual composable representative subspace	Flat

Result	Exact declaration	Assumptions	Model
Discrete full-word quotient	`discrete_traceSensitiveQuotient`	Fixed endpoints and discrete labels; alphabet need not be finite	Flat

Table 3: Presentation invariance and separation verification map. Names are under `ComputationalPaths.Path.GeometricTopology`. Every row has source commit `0346c8e`.

Result	Exact declaration	Assumptions	Model
Word substitution	`FlatWordSubstitution.continuous_substitution`	Continuous primitive word codes; arbitrary finite output lengths	Flat
Fixed invariance	`ScopedWordInterpretation.flatSensitiveQuotientHomeomorph`	Rule preservation; primitive roundtrip rewrites; continuous word codes	Flat
Global invariance	`ScopedWordInterpretation.flatSensitiveGlobalHomeomorph`	Same hypotheses; common ambient space	Flat
Final-pair invariance	`ScopedWordInterpretation.flatSensitiveFinalPairHomeomorph`	Same hypotheses; actual matching-endpoint representative subspace	Flat
Projection compatibility	`ScopedWordInterpretation.finalPairMap_left`, `ScopedWordInterpretation.finalPairMap_right`	No ordinary product-quotient hypothesis	Flat
Abbreviation invariance	`RedundantGeneratorObservable.flatSensitiveGlobalHomeomorph`, `RedundantGeneratorObservable.flatSensitiveFinalPairHomeomorph`	Discrete old and new alphabets; the actual defining rule	Flat
Timing nonhomeomorphism	`FlatObservableDiscreteness.BinaryCircle.binary_flat_not_homeomorphic`	Actual three-label circle paths; fixed endpoints; no named rules	Both
Complete presentation	`CompleteDiscreteSeparation.complete`	All continuous paths as discrete labels; homotopy named rules	Both
Ordinary quotient recovery	`CompleteDiscreteSeparation.observableHomeomorph`	Any based topological space; compact-open path topology	Flat
Complete separation	`CompleteDiscreteSeparation.not_homeomorphic`	Ordinary based quotient explicitly assumed nondiscrete	Flat

8 October 2026: all-loop coverage. The current direct-replay manifest selects 178 actual declarations (127 theorems and 51 definitions), extending the 141-declaration represented-rose selection. Its source and dependency closure are audited against the manifest before NanoDa replay. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. The exact published-head Quality run checks these declarations separately from the five retained Comparator selections. The rose development adds an actual geometric carrier and a checked Cayley-edge covering to the reduced-word calculation. It proves completeness between represented traces, geometric noncommutation, and discreteness of the two fixed-endpoint flat computational quotients. The all-loop extension constructs an actual continuous contraction

of that covering realization, derives simple connectivity, and classifies arbitrary continuous rose loops by reduced words. Its multiplication convention and realization comparison are explicit in Table 5.

Table 4: Geometric rose verification map. Every name has prefix `ComputationalPaths.Path.GeometricTopology`. The source snapshot is `dbfe65e`, with Lean 4.32.0 and Mathlib `81a5d257`. The listed results use only the three permitted standard axioms.

Result	Exact declaration	Assumptions	Model
Actual rose	`GeometricRose.axesHomeomorph`	Two additive circles, exactly their zero points glued	Geometric
Reduced normal form	`RoseReducedWords.normalization`, `RoseReducedWords.scoped_iff_encode_eq`	Any two based primitive loops; no named rewrites	Scoped algebra
Geometric covering	`CayleyRoseCovering.covering`	Actual interval-glued free-group edges; proved quotient and orbit fibers	Geometric
Computed endpoint	`RoseLiftedWords.liftPath_trace_one`	Actual rose primitives and covering; identity starting vertex	Binary path
Trace completeness	`RoseLiftedWords.trace_complete`, `RoseLiftedWords.flat_trace_complete`	Based finite traces; actual endpoint-fixed homotopy	Binary and flat
Order separation	`RoseLiftedWords.ab_not_homotopic_ba`	Actual once-around primitive loops	Geometric
Signed observation	`RoseBasedTopology.signedPrimitive_injective`	The four actual oriented paths; winding pairs through retractions	Parametrized
Based quotients	`RoseBasedTopology.flatObservableHomeomorph`, `RoseBasedTopology.flatSensitiveHomeomorph`	Fixed endpoints; respective proved discrete quotient topologies	Flat
Tree prefix step	`CayleyReducedPrefix.outward_word_cases`	Actual reduced free-group words; no geometric assumptions	Algebraic

`CompleteDiscreteSeparation` checks the abstract mechanism of Proposition 2.6. Its labels are all continuous interval paths, with a separate discrete label topology, and its named rules are exactly homotopies of parallel realized traces. The global geometric-completeness criterion is proved directly. For based loops at a fixed x, `observableHomeomorph` identifies the flat observable quotient with the ordinary quotient fundamental group, while `sensitive_discrete` proves discreteness of the full-word quotient for the possibly infinite alphabet. With nondiscreteness of the ordinary based quotient as an explicit hypothesis, `not_homeomorphic` excludes every homeomorphism between the two topologies. Matching coherent-representative versions are checked too. The harmonic archipelago and Fabel's facts about its based quotient remain external mathematical inputs; no such facts are introduced as Lean axioms.

Two Lean developments accompany the paper. The parent artifact (Section S4.1) uses Lean 4.24.0 and Mathlib 4.24.0 and covers the constructions listed in Table 7.

A smaller extraction, built with Lean 4.32.0, contains one bundled statement registered in Palomar (Section S4.2). The statements and proofs in this paper can be read without either implementation, and in both cases the checks cover the named declarations and nothing else.

Table 5: All-loop rose verification map. Names have prefix `ComputationalPaths.Path.GeometricTopology`. The source snapshot is `3fd21d0`; Lean 4.32.0, Mathlib `81a5d257`; only the three permitted standard axioms.

Result	Exact declaration	Assumptions	Model
Exact edge timing	`CayleyFlatWalk.walkPath_prefix`, `CayleyFlatWalk.outward_last_edge`, `CayleyFlatWalk.inward_last_edge`	Actual signed geometric edges; reduced-word prefix proved	Flat
Joint contraction	`CayleyContraction.continuous_contract`, `CayleyContraction.homotopy`	Actual closed-edge quotient; local compactness of the compact Hausdorff interval	Geometric tree
Simple connectivity	`CayleyContraction.contractible`, `CayleyContraction.simplyConnected`	Constructed homotopy; no simple-connectivity input	Geometric tree
Full fundamental group	`RoseFundamentalGroup.fundamentalGroupEquiv`, `RoseFundamentalGroup.freeGroupEquiv`	Concrete covering; proved simple connectivity	Opposite F_2, then inversion
Every continuous loop	`RoseFundamentalGroup.loop_homotopic_flat_decode`, `RoseFundamentalGroup.loopWord_eq_iff_homotopic`	Any actual based continuous loop	Flat; endpoint-fixed homotopy
Realization comparison	`RoseFundamentalGroup.traceQuotientEquiv_flat_mk`, `RoseFundamentalGroup.traceQuotientEquiv_mul`	Structural scoped rules; explicit opposite multiplication	Flat; algebraic quotient

S4.1. **2026 parent archive: Lean 4.24.0.**
Traces, soundness, and quotient operations. The implementation represents traces by the parenthesized inductive type `GeometricTrace`, where Section 2 uses flat signed words. The declarations `WeightedSlotReparam`, `balancedSlotReparam`, and `weightedConcatenationInvariant` relate the parametrizations up to endpoint-fixed homotopy, including the null-homotopy assumptions needed when a factor has zero weight. They do not identify observable topologies. The scoped-rewrite modules check soundness and the quotient operations. The comparison with Mathlib's fundamental groupoid identifies the represented homotopy classes, checks preservation of identities, inverses, and composition, and proves that the induced map on final composable-pair domains is continuous.
Representative topologies. The module `TraceSensitiveTopologicalCompPath` flattens internal traces to signed words while retaining binary realization in its observable coordinate. It checks continuity of that binary realization, of the map to observable coordinates, and of the identity, reversal, and binary composition

operations. The earlier quotient modules still use the observable topology by default.

The module `TraceSensitiveSeparation` checks the two-label separation mechanism on a finite code space: the one-letter trace codes differ, their observable codes agree, and the identity from the discrete trace topology to the indiscrete observable topology is continuous while its inverse is not. It also checks that the two circle labels realize the same nonconstant loop, proves the signed-count invariant for every structural scoped rewrite, and separates their scoped quotient classes. Their full observable codes agree. Signed count descends continuously to the trace-sensitive quotient; the comparison to the observable quotient is continuous, but its inverse is not. Thus the topological separation in Example 2.8 is checked in Lean. The module `TraceSensitiveUniversalCollapse` checks the continuous one-letter section and the homeomorphism between the two universal quotient topologies.

Circle and torus examples. The module `FiniteCircleTorusPresentation` uses one circle generator and two torus generators. It checks integer and integer-pair codes for traces, completion certificates, and soundness of the torus commuting square. The modules `BasedScopedTraceGroup`, `FiniteCircleScopedCompleteness`, and `FiniteTorusScopedCompleteness` prove normalization for the exact finite presentations. The circle proof uses only the structural groupoid laws; the torus proof adds its single named commuting square. Both prove based geometric completeness, instantiate the trace-topology comparison, and identify the based scoped quotients with discrete $\mathbb{Z}$ and $\mathbb{Z}^2$, respectively. Composing with the geometric winding homeomorphisms gives explicit homeomorphisms from the scoped based quotients to the ordinary circle and torus loop quotients. The separate torus development in `ComputationalPaths/Path/Topology/TopologicalTorusScoped.lean` includes the comparison between simultaneous and sequential loop representatives.

Hawaiian-earring transfer. The dedicated module checks that an externally supplied failure of the product-quotient property yields failure of ordinary-pair compatibility, and it transfers discontinuity of multiplication along the comparison homeomorphism. Fabel's results enter as assumptions; the Lean development does not reprove them.

Build and archive. The root import file is `ComputationalPaths.lean`, and Table 7 in Appendix S5 lists the principal declarations. The core artifact, version `0.6.1+lean-only-v3`, is archived on Zenodo under the version DOI `doi:10.5281/zenodo.21938980`; its concept DOI is `doi:10.5281/zenodo.21817207` [5]. The source, including the finite-generator module, the torus simultaneous/sequential bridge, and the trace-sensitive topology module, is also in the GitHub repository. Build the project with `lake build`; the artifact README lists commands for building individual modules separately.

The checked modules contain no unfinished proofs and no custom axiom declarations. Declaration-level axiom inspection reports only Lean's standard logical and quotient principles: propositional extensionality, classical choice, and quotient soundness. The development complements earlier work on computational paths and the Lean theorem prover [1, 3, 4].

S4.2. **29 August 2026: registered version 1.** The extraction *TopologicalComputationalPaths* has a registered Palomar result, `PALOMAR-2026-08-29-000005`, version 1, published on 29 August 2026 [6]. It selects the single bundled declaration

`TopologicalComputationalPaths.main_result` at the immutable commit

`8254c40d0de03ff469c7c9c05087b8bc154e9a87`,

built with Lean 4.32.0 and its checked-in dependency manifest. The selected statement is an `OrdinaryTopologyComparisonCertificate`, and Table 6 summarizes what it covers.

The certificate has three parts. The first gives the continuous bijection between final and ordinary pair spaces, the equivalent criteria for their topologies to agree, the compact-Hausdorff and discrete sufficient conditions, the consequence for ordinary composition, and the obstruction obtained from discontinuity. (The discrete field assumes that both $G_{\mathcal{P}}$ and its final domain are discrete; by Corollary 4.8 the first assumption implies the second.) The second part classifies based-loop homotopy classes of the circle and torus by integers and integer pairs. It checks the invariant, the standard representatives, the identity value, additivity under concatenation, and that the classification maps are mutually inverse. The homeomorphism statements of Theorems S3.1 and S3.2 are not part of it. The third part transfers the Hawaiian-earring obstruction under the hypotheses described below.

Part of the paper	Scope of the registered certificate
Final/ordinary comparison	Canonical continuous bijection; exact quotient/homeomorphism/topology criteria; sufficient conditions and ordinary-composition consequence.
Circle and torus	Additive classifications of the actual loop quotients, with explicit standard representatives and inverse laws.
Hawaiian earring	Concrete observable based fiber and quotient comparison; obstruction transfer conditional on `FabelHawaiianEarringFacts`.
Broader exposition	Not certified by this selection: the trace-sensitive refinement, the minimal finite-generator rewrite algorithms, the homeomorphism statements for the circle and torus, the explanatory homotopies and figures, and the manuscript as a whole.

TABLE 6. What the Palomar registration covers. Supporting code and mathematical exposition are listed apart from the selected conclusions.

The Hawaiian-earring field needs a caveat. In the registered statement, the topology `hawaiianObservableOpenFiberTopology` is induced by the geometric projection alone, and its one-letter section identifies its quotient with the standard loop quotient. This topology was defined for the registered based-fiber result; identifying it with the topologies of Section 2 would need a proof. The field `hawaiian_based_fiber` assumes `FabelHawaiianEarringFacts`, so the failure of the product-quotient property and the discontinuity of multiplication are hypotheses there, and the extraction does not reprove them. In the mathematical account, Proposition 4.9 takes them from Fabel's published result [2].

24 September 2026: working-source extensions. The later working-source milestone equipped the Hawaiian observable based carrier with the topology induced by its full observable code. The revised one-letter section and geometric projection were

checked continuous, with the based quotient homeomorphic to the usual loop quotient. That milestone also weakened the discrete sufficient condition in the selected certificate to discreteness of the arrow space alone, and checked continuity of both projections from the final composable domain. It checked the general open-arrow criterion and Proposition 7.5: the path-class projection is quotient, and its openness implies ordinary multiplication for the universal presentation. A direct Lean proof supplied openness under the hypotheses of Theorem 7.6. The theorem itself remains credited to Holkar, Hossain, and Kulkarni.

On fixed endpoint quotients, Lean identified the universal based fiber with the ordinary based-loop quotient. It proves discreteness, a quotient pair map, and continuous ordinary multiplication under the semilocal hypothesis. It also identifies based paths with the endpoint-defined subspace of the compact-open path space and checks that an open global projection restricts to an open quotient onto the based-arrow subspace. For locally path-connected, semilocally simply connected spaces, the checked open-map theorem removes the remaining hypothesis from that comparison. The resulting global based-arrow subspace is homeomorphic to the ordinary based-loop quotient and is discrete. The module `UniversalTotallyDisconnected` proves the unconditional case of Corollary 7.7. Its projection is a homeomorphism, and ordinary composition is continuous. `EndpointVaryingLadder` establishes the local ladder step: every path has a compact-open neighborhood whose members differ by short endpoint connectors. `EndpointAbsorption` first gives the center path positive constant tails. It then inserts connectors confined to suitable endpoint neighborhoods and proves, by an explicit reparametrization, that the new path has the required concatenation homotopy class. The module `UniversalSemilocallySimplyConnected` uses both lemmas to show that the saturation of any compact-open set is open. It also records ordinary pair compatibility, continuous composition, and the global based-fiber homeomorphism as consequences. The integer-indexed scoped circle presentation has a checked normal form; its fixed endpoint quotient is homeomorphic to $\mathbb{Z}$, and a continuous winding-based trace choice identifies its trace-sensitive and observable quotient topologies. This integer-indexed certificate is distinct from the finite-generator circle presentation in Section 8. That working source also checked the general trace-section criterion, the universal trace collapse, a finite separation model, and the realized fundamental-groupoid arrow comparison. For products it defines the horizontal and vertical continuous step system, checks geometric soundness of primitive interchange, and proves sorting and based trace completeness for the presentation in Theorem S3.3. The module `ProductPrimitiveInterchange` derives the whole-trace rule from primitive rectangles, shows that the two presentations induce the same scoped rewrite relation, and transfers based completeness to the smaller presentation. For the finite-generator circle and torus, it checks the continuous winding-based choices of signed traces and proves the required based completeness for their exact scoped rules. The resulting based quotient homeomorphisms and trace-topology comparisons are unconditional. A comparison with the subspace fibers of their global arrow quotients is also checked: translation of a variable-basepoint loop to zero gives a continuous winding map on each global scoped quotient, and finite based completeness makes its restriction to the actual based-arrow subspace injective. The resulting subspaces are homeomorphic to $\mathbb{Z}$ and $\mathbb{Z}^2$, respectively, and thus to the ordinary geometric loop quotients. The

immutable source for that milestone is `TopologicalComputationalPaths` commit `09b20abe0a4f69b361846c2466bff8528044fa93`. With the repository's Lean 4.32.0 toolchain and checked-in Mathlib manifest, `lake build` checks the development and all configured targets. The manuscript source, bibliography, and figures are in `paper/topological` and `paper/refs.bib`; the manuscript is not itself a theorem selected by Comparator. Those working-source additions had not been registered as a new Palomar version. The internal `comparator-roadmap.json` bundle selects a 24-field certificate containing the checked semilocal and totally disconnected universal cases, product results, finite global fibers, and trace-topology comparisons. Its Challenge imports the substantive development, and its Solution assembles the proved declarations. Comparator, NanoDa, and Lean's kernel replayed this bundle in CI. The Challenge provenance rule at that milestone excluded local imports, so the bundle was not a registration candidate. A separate Mathlib-only Challenge in `comparator-registry-roadmap.json` selects the open-arrow ordinary-pair quotient, continuity transfer for multiplication, and the ordinary geometric circle and torus loop-quotient homeomorphisms. At that milestone, these four statements had not undergone Palomar's mechanical preflight or external review. The finite scoped global-fiber results were outside that selection.

To reproduce the registered version 1 result, check out its pinned commit and run `lake build`. The selection file `comparator.json` pairs `Challenge.lean` with `Solution.lean`. The theorem placeholder in `Challenge.lean` states what is checked against the proof in `Solution.lean`, so it does not count as an unfinished proof. Later follow-up and preimage-solver development is outside this registered snapshot. Registration gives a versioned verification record for the selected formal statement and nothing more. It does not cover the prose or the illustrations, and it implies neither journal acceptance nor an endorsement of novelty.

S5. Declaration index for the parent artifact

This index records the representative declarations discussed in Section S4.1. It describes the parent Lean artifact, not the independently registered Palomar selection of Section S4.2. Fully qualified names are retained for source lookup.

Table 7: Representative declarations in the parent artifact, not the Palomar selection.

Mathematical layer	Representative checked declarations
Scoped syntax and soundness	`ScopedGeometricRewritePresentation`, `ScopedRwEq`, `ScopedRwEq.sound`
Weighted realization	`WeightedSlotReparam`, `balancedSlotReparam`, `weightedConcatenationInvariant`
Trace-sensitive topology	`TraceSensitiveTopologicalCompPath.lean`, `continuous_fullTraceRealization`, `continuous_traceSensitive_to_observable`, `traceSensitiveTopologyCertificate`, `TraceSensitiveQuotient.continuous_quotientComparison`, `TraceSensitiveQuotient.quotientComparisonHomeomorph_of_realization_section`

Table 7 continued

Mathematical layer	Representative checked declarations
Finite topology separation	`TraceSensitiveSeparation.oneLetter_e_ne_f`, `TraceSensitiveSeparation.observableCode_e_eq_f`, `TraceSensitiveSeparation.not_continuous_observable_to_trace`, `TraceSensitiveSeparation.certificate`
Quotient groupoid and topology	`ScopedComposableClass`, `scopedCompositionOnStrong`, `scopedFinalTopologicalGroupoidCertificate`
Ordinary/final comparison	`scopedProductCompatibility_iff_four_way`, `scopedProductCompatibility_of_compact_final_t2`
Comparison and normal forms	`ComparisonFunctorCertificate`, `comparisonHomeomorph_of_complete`, `ScopedGeometricNormalFormCertificate`
Universal and obstruction cases	`universalHomeomorph`, `ScopedGeometricRewrite.universalTraceSensitiveHomeomorph`, `QuotientObstructionTransfer`, `MultiplicationComparison`
Finite-generator presentation data	`FiniteCircleTorusPresentation.circleFinitePresentation`, `FiniteCircleTorusPresentation.circleCompletionEquivInt`, `FiniteCircleTorusPresentation.torusFinitePresentation`, `FiniteCircleTorusPresentation.torusCompletionEquivIntProd`, `FiniteCircleTorusPresentation.torusFiniteCommutingSquare`
Completed circle and topological torus	`circleBasedNormalFormCertificate`, `TopologicalTorus.standardLoop_homotopic_sequentialLoop`, `TopologicalTorus.equivIntProd`, `TopologicalTorus.certificate`

References

1. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Proceedings of the 28th International Conference on Automated Deduction (CADE-28), Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
2. Paul Fabel, *Multiplication is discontinuous in the Hawaiian Earring Group (with the quotient topology)*, Bulletin of the Polish Academy of Sciences. Mathematics **59** (2011), no. 1, 77–83.
3. Arthur F. Ramos, Anjolina G. de Oliveira, Ruy J. G. B. de Queiroz, and Tiago M. L. de Veras, *Formalizing computational paths and fundamental groups in Lean*, 2025, arXiv preprint arXiv:2511.19142.
4. Arthur F. Ramos, Tiago M. L. de Veras, Ruy J. G. B. de Queiroz, and Anjolina G. de Oliveira, *Computational paths form a weak ω-groupoid*, 2025, arXiv preprint arXiv:2512.00657.
5. Arthur Freitas Ramos, Ruy J. G. B. de Queiroz, Anjolina Grisi de Oliveira, and Tiago M. L. de Veras, *Topological semantics for scoped computational paths—lean artifact*, Zenodo software archive, 2026, Version 0.6.1+lean-only-v3, doi:10.5281/zenodo.21938980.
6. Arthur Freitas Ramos, Ruy J. G. B. de Queiroz, Anjolina Grisi de Oliveira, Tiago M. L. de Veras, and David Barros Hulak, *TopologicalComputationalPaths: registered lean result*, Palomar, PALOMAR-2026-08-29-000005, version 1, 2026, Registered 29 August 2026. Source commit 8254c40d0de03ff469c7c9c05087b8bc154e9a87.